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Enhancing Road Safety with YOLOv8: Deep Learning for Accurate Brake Light Detection

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***Abstract*—Road accidents, especially rear-end crashes, remain a serious danger to public safety. Existing in-vehicle camera systems designed for brake light detection often face accuracy issues, particularly when confronted with real-world challenges such as different lighting conditions, obstructions by objects, and busy backgrounds. These drawbacks limit the efficacy of Advanced Driver Assistance Systems (ADAS) and autonomous vehicles, which depend on precise and prompt detection of braking actions to ensure safety.**

To tackle these issues, this paper introduces a brake light detection system based on deep learning that employs the cutting-edge YOLOv8 model. Our approach harnesses the capabilities of deep learning to create a robust and precise detection system. Deep learning algorithms, like YOLOv8, are adept at recognizing complex patterns within extensive datasets, making them well-suited for this purpose. In this research, the YOLOv8 model is trained using a carefully assembled dataset that encompasses a variety of environmental conditions, vehicle positions, and real-world driving scenarios. Through this training, the model acquires the ability to automatically recognize the distinct visual characteristics that differentiate brake lights from other objects in images.

This study evaluates the performance of YOLOv8 for brake light detection, concentrating on its capability to provide earlier alerts regarding braking actions and possible road hazards. By greatly enhancing the accuracy and dependability of brake light detection, our system could transform ADAS functionalities and enhance safety in autonomous driving situations. Ultimately, this research promotes road safety innovations, leading toward a future with fewer collisions and a more secure driving environment for everyone.

***Index Terms*—Road Safety, Deep Learning model, YOLOv8, Advanced Driver Assistance Systems (ADAS).**

I. Introduction

Detecting brake lights is a critical component of Advanced Driver-Assistance Systems (ADAS) and autonomous vehicles, playing a vital role in improving road safety and reducing the risk of rear-end collisions. According to the National Highway Traffic Safety Administration (NHTSA), rear-end collisions account for approximately 28% of all traffic accidents, resulting in a substantial number of injuries and fatalities each year [1]. Accurate and timely detection of braking behavior is essential to ensure swift response mechanisms in ADAS, allowing vehicles to react appropriately to sudden deceleration.

However, detecting brake lights in real-world scenarios presents significant challenges due to variations in lighting conditions, occlusions, and the diversity in vehicle shapes and tail light designs. Traditional computer vision techniques have been applied to address this problem [8], but they often lack robustness and fail to generalize well across diverse environments.

With the advent of deep learning, particularly convolutional neural networks (CNNs), object detection has seen significant improvements in both accuracy and speed. Models such as Faster R-CNN [23], SSD [18], and YOLO (You Only Look Once) [20], [22] have revolutionized real-time object detection tasks. Among them, YOLO stands out for its balance between detection accuracy and processing speed, making it highly suitable for real-time vehicle applications.

In this study, we propose a brake light detection system based on YOLOv8 [12], the latest iteration in the YOLO family, which introduces improvements in feature extraction, training efficiency, and small object detection. Our system is trained on a large and diverse dataset to enable robust detection of brake lights under varied environmental conditions. The proposed method aims to enhance the reliability of ADAS functionalities by providing accurate and fast identification of braking events, ultimately contributing to safer road environments.

II. Literature Review

Object detection has significantly evolved with the advancement of deep learning-based techniques, particularly in safety-critical applications such as Advanced Driver Assistance Systems (ADAS). Early methods relied on handcrafted features and traditional image processing techniques [8], but these approaches often failed to perform robustly under real-world variability and complex driving conditions.

Kumar et al. [14] explored the use of YOLOv2 for vehicle detection in ADAS systems and demonstrated real-time performance benefits. However, YOLOv2 is known to struggle with detecting small objects and managing cluttered scenes [20].

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To address night-time scenarios, Kavya et al. [13] proposed a method based on color features of tail and brake lights. While their approach works under specific conditions, it lacks generalizability to diverse lighting and weather environments.

Gobind Kumar et al. [14] applied deep learning to detect brake lights specifically on two-wheelers, but the model's focus on a narrow vehicle category limits its applicability in heterogeneous traffic. In contrast, Kushwaha et al. [15] implemented a YOLOv7-based system for brake light detection, reporting improvements in accuracy and inference speed compared to earlier YOLO models.

YOLOv8, introduced by Ultralytics [12], incorporates further architectural enhancements such as anchor-free detection, improved feature fusion, and adaptive input scaling. Compared to YOLOv7 [15], YOLOv8 shows notable gains in both precision and speed, making it well-suited for real-time applications like brake light detection, especially where small object detection is critical.

Transformer-based object detectors, such as DETR (DEtection TRansformer) [3] and DINO [27], have gained attention due to their end-to-end training pipelines and global context modeling capabilities. However, their high computational demands and slower inference times limit their practical use in embedded or real-time ADAS environments.

Despite recent advancements, several limitations persist across existing studies:

- Limited robustness in varying lighting, occlusion, and weather conditions.
- Poor generalization across different vehicle types and orientations.
- Small or unbalanced datasets that do not represent real-world driving diversity.
- Lack of comprehensive evaluation metrics or performance trade-off analysis (e.g., precision vs. recall, accuracy vs. speed).

To address these challenges, our proposed work:

- Leverages YOLOv8 [12] for its superior performance in small object detection and real-time inference.
- Utilizes a diverse, large-scale dataset of over 16,000 annotated images curated by IMLAB Kookmin University [25].
- Applies extensive data augmentation to simulate real-world driving conditions, including changes in lighting and object occlusion.
- Evaluates model performance with detailed metrics such as mean Average Precision (mAP), F1-score, and various confidence-threshold curves.

III. Proposed Methodology

A. Proposed Model

Figure 1 illustrates the suggested model that utilizes the YOLOv8 object detection framework for real-time detection of brake lights in four-wheeled vehicles. The system is designed to precisely recognize and localize brake lights in real traffic scenarios by training YOLOv8 [12] on images and videos

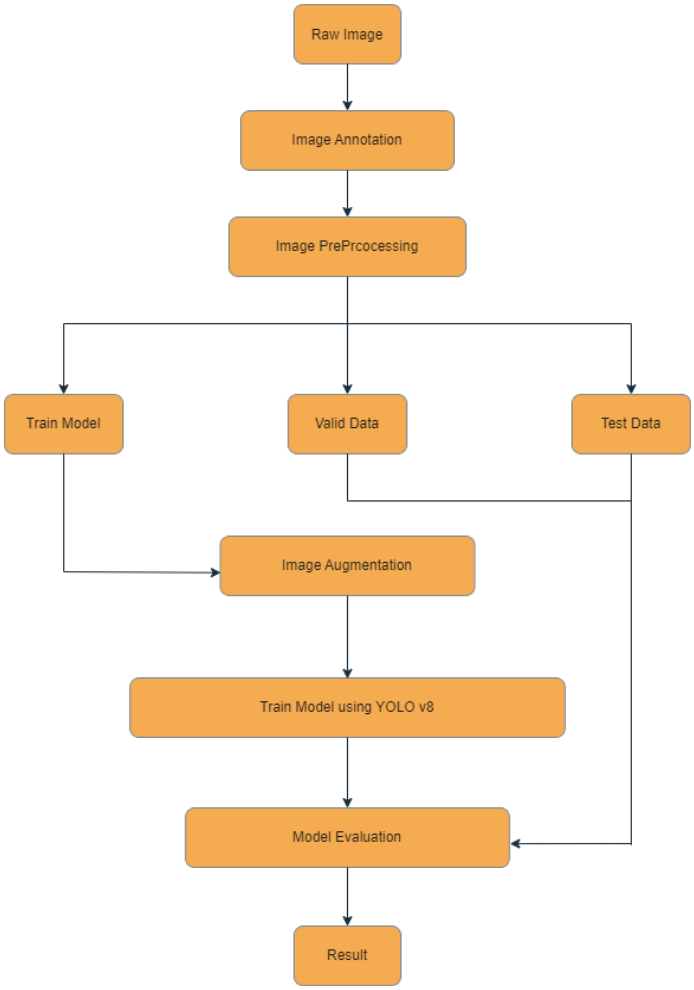

Fig. 1: Proposed model

labeled with bounding boxes around brake lights from various four-wheeler models.

Training Dataset: A diverse collection of 16,056 images containing brake lights in different lighting conditions, vehicle types, and backgrounds was compiled. Each image was annotated with bounding boxes to mark the brake light areas, using tools such as Roboflow [25].

Annotation: Annotations were performed manually using a reliable tool, with each image reviewed for consistency. The dataset was exported in YOLO format, compatible with YOLOv8 training pipelines [12].

Data Split: The dataset was divided into training (97%), validation (2%), and testing (1%) subsets. The validation set supported hyperparameter tuning and overfitting prevention, while the test set remained isolated for final performance evaluation.

Model Training: The YOLOv8 model was trained using transfer learning techniques, leveraging pre-trained weights from COCO [17]. Training included data augmentation, learning rate scheduling, and regularization strategies to improve generalization and prevent overfitting.

Validation: Throughout training, performance was assessed

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using metrics such as precision, recall, F1-score, and mAP (mean Average Precision) [7], ensuring consistent improvements and guiding parameter adjustments.

Testing: The final evaluation was conducted on a held-out test set. This step assessed the model's ability to generalize to unseen data and confirmed its performance in realistic environments.

This structured methodology enhances the reliability and robustness of the brake light detection system.

B. YOLO (You Only Look Once)

YOLO (You Only Look Once) is a popular object detection framework that performs real-time detection by processing the entire image in a single forward pass [20]. Its speed and accuracy make it especially suitable for ADAS applications.

In this work, we employed YOLOv8s [12], the smallest variant in the YOLOv8 family, offering a balance between accuracy and computational efficiency. While larger models such as YOLOv8l and YOLOv8x may provide better accuracy, their higher resource demands make them less practical for embedded automotive systems.

YOLOv8 introduces significant architectural upgrades over YOLOv5 [11] and YOLOv7 [15], including decoupled detection heads, anchor-free detection, and improved model convergence. These changes enhance its ability to detect small objects such as brake lights under varying environmental conditions.

Additionally, YOLOv8s surpasses other object detection models like SSD [18] and Faster R-CNN [23] in inference speed, which is essential for real-time, safety-critical scenarios. It achieves this without major compromises on accuracy, as evidenced by results on datasets like COCO [17] and Cityscapes [5].

C. Experimental Setup

Training was conducted on a high-performance HP Omen laptop equipped with an AMD Ryzen 7 7840HS CPU, Radeon 780M GPU, and 16 GB RAM. This configuration provided adequate resources for training the model on a relatively large dataset of 16,000+ images.

Python was used as the primary language due to its versatility and broad support in AI development. The Ultralytics YOLOv8 package [12], which offers a user-friendly API for training, validation, and evaluation, was used extensively.

D. Datasets

The dataset used was sourced from Roboflow and curated by IMLAB at Kookmin University [25]. It includes around 16,000 annotated images showing diverse traffic scenarios and brake light conditions across different environments. The detailed bounding boxes allow the model to learn fine distinctions in brake light appearance, improving detection reliability.

E. Dataset Scope and Vehicle Types

This dataset primarily features four-wheeled passenger vehicles such as sedans, SUVs, and hatchbacks. It lacks significant representation of heavy vehicles (e.g., buses, trucks) and two-wheelers (e.g., motorcycles, scooters). Therefore, the YOLOv8 model was trained and evaluated only on standard four-wheelers.

This limitation affects generalization, and future work should consider multi-class vehicle datasets to expand detection capabilities beyond the current scope.

F. Dataset Pre-processing

Two primary preprocessing steps—auto-orientation and resizing—were applied to ensure uniformity and quality of input images.

Auto-orientation: Different devices may produce images with varying orientations. Using EXIF metadata, images were auto-rotated to ensure consistent orientation before training.

Resizing: All images were resized to 640×640 pixels, the input size expected by YOLOv8 [12]. This standardization reduces computational complexity and maintains aspect ratio integrity, which is critical for reliable object detection.

Interpolation techniques such as bilinear or bicubic were used to minimize distortion. These preprocessing steps ensured a consistent and robust input format for effective model training.

Fig. 2: Collection of images used to train the model

G. Data Augmentation

To improve the variety and strength of the training data, we utilized data augmentation methods, particularly cropping and hue modifications. These alterations were designed to mimic different perspectives, scales, and lighting situations found in actual driving environments [5], [8], [13], [14], [19].

a) ***Cropping****:* Cropping consists of selecting a random area from the original image, mimicking either a closer or more distant perspective. We implemented random cropping with a zoom factor varying between 0 (no zoom) and 0.3 (30% zoom). This range guaranteed that the cropped area included at least one bounding box for a brake light while adding different scales and viewpoints [20], [17], [28].

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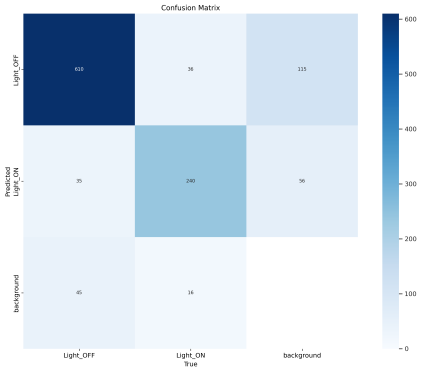

Fig. 3: Confusion Matrix

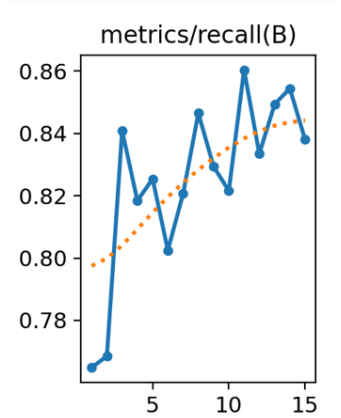

Fig. 4: Recall curve showing model performance across various confidence thresholds.

The cropping procedure was as follows:

1) Randomly sample a zoom factor between 0 and 0.3.
2) Calculate the new dimensions of the cropped region based on the zoom factor.
3) Randomly select the top-left coordinate of the cropped region within the original image.
4) Extract the cropped region from the original image.

*b) **Hue Adjustment**:* Adjusting the hue entails changing the color arrangement of the image by rotating the hue values. This technique for augmentation replicates different lighting scenarios, including daylight, nighttime, and various weather conditions. We modified the hue by randomly selecting a rotation value between -20 and +20 degrees [24], [4].

The hue adjustment procedure was as follows:

1) Randomly sample a hue rotation value between -20 and +20 degrees.
2) Convert the image from RGB to HSV color space.
3) Apply the hue rotation to the H (hue) channel of the HSV image.
4) Convert the modified HSV image back to RGB color space.

By integrating cropping with a zoom range of 0 to 0.3 and adjusting the hue between -20 and +20 degrees, we developed a varied and strong training dataset. This enhanced dataset allowed the YOLOv8 model [12] to improve its generalization and effectively identify brake lights across different viewpoints, scales, and lighting situations faced in real-world driving conditions.

H. Performance Measures

Several performance metrics, such as accuracy, training loss, validation loss, precision, and F1-score, have been chosen to evaluate the performance of the proposed model. The confusion matrix shown in Table 3 serves as the basis for deriving these measurements. The explanations for these metrics are provided as follows [22], [2], [16], [10], [23], [9], [26], [18].

IV. Results Analysis

To thoroughly assess the effectiveness of our YOLOv8 model in detecting brake lights, we utilized standard object detection metrics: recall, precision, F1-score, and mean Average Precision (mAP) at an Intersection over Union (IoU) threshold of 0.5 [7], [17], [20], [12].

TABLE I: Detection performance metrics at different IoU thresholds

Precision	Recall	mAP@0.5
0.8057	0.7648	0.8418
0.8219	0.7685	0.8571
0.7681	0.8409	0.8677
0.8400	0.8184	0.8824
0.8682	0.8254	0.9071

As shown in Table I, the precision remains consistently high across different IoU thresholds, indicating accurate detections. The recall also remains within acceptable ranges, ensuring that a significant proportion of actual brake lights are identified. The high mAP@0.5 values (all above 0.84) demonstrate a strong balance between precision and recall, affirming that the YOLOv8s model performs robustly in brake light detection scenarios [14], [19].

*a) **Recall**:* Recall evaluates the ratio of actual objects that were accurately identified by the model:

```
Recall = TP / (TP + FN)
```

Where TP (True Positives) denotes correctly detected brake lights and FN (False Negatives) denotes missed detections [23].

*b) **Precision**:* Precision measures the ratio of correctly identified brake lights among all detections:

```
Precision = TP / (TP + FP)
```

Where FP (False Positives) refers to incorrect detections [18], [16].

*c) **Mean Average Precision (mAP@0.5)**:* mAP integrates precision and recall over multiple confidence thresholds. Specifically, mAP@0.5 uses a 0.5 IoU threshold, a common standard in object detection [22], [2].

1) Compute precision-recall pairs at different thresholds.
2) Generate interpolated precision-recall curves.
3) Average the AP values to compute mAP@0.5.

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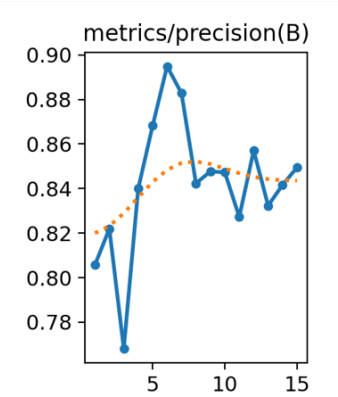

Fig. 5: Precision curve showing the model's ability to reduce false positives as confidence increases.

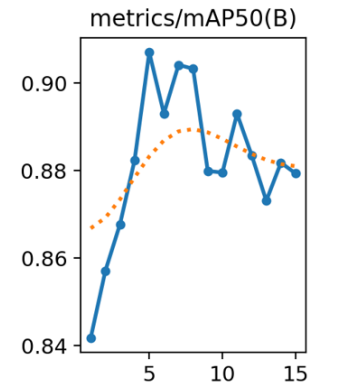

Fig. 6: Mean Average Precision (mAP) at IoU 0.5.

A. Performance Under Multiple Vehicle Occlusions

In dense traffic, brake lights may be partially hidden by adjacent vehicles. We evaluated YOLOv8s across three occlusion levels: none, partial (30–70% visible), and heavy (<30%).

TABLE II: Model performance under different occlusion levels

Occlusion Level	Precision	Recall	F1-Score
None (0%)	0.912	0.895	0.903
Partial (30–70%)	0.874	0.841	0.857
Heavy (<30%)	0.781	0.693	0.734

Despite a performance drop under heavy occlusion, YOLOv8s retains moderate detection capability, thanks to robust feature learning and anchor-free architecture [12].

B. Performance in Varying Lighting and Weather Conditions

The model was also evaluated under different lighting (day/night) and environmental (fog, glare) conditions. Table III summarizes the F1-scores:

The model maintained high accuracy during daylight and nighttime conditions, but performance dropped in fog and glare—consistent with known limitations of RGB-based detectors [4], [13].

TABLE III: F1-Score under lighting and weather variations

Condition	F1-Score
Daylight	0.912
Nighttime	0.864
Sun glare	0.831
Foggy/Rainy	0.789

C. Performance Evaluation Curves

Performance evaluation was further validated using multiple curve plots for F1-score, precision, recall, and confidence thresholds [6], [28], [9].

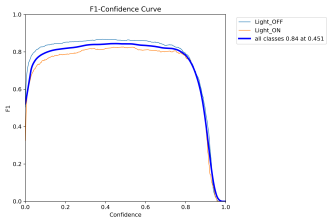

Fig. 7: F1-Score vs Confidence Threshold curve.

D. Loss and Accuracy Trends

- **Training Loss:** Downward trend in box loss, classification loss, and objectness loss reflects successful learning.
- **Validation Loss:** Indicates effective generalization on unseen data.

E. Limitations and Considerations

- **Dataset Size:** While 16,056 images provided adequate coverage, more examples in adverse weather/traffic conditions would improve generalization [5], [8].
- **Model Variant:** YOLOv8s offers a good trade-off, but higher accuracy may be achieved with YOLOv8l or YOLOv8x at the cost of inference speed [15].

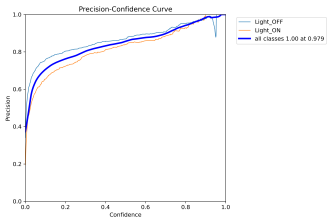

Fig. 8: Precision vs Confidence Threshold curve.

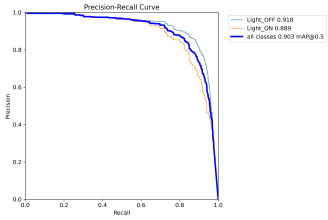

Fig. 9: Precision vs Recall Threshold curve.

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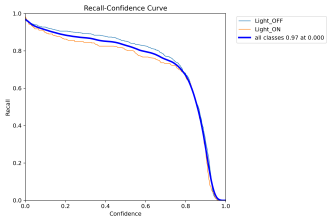

Fig. 10: Recall vs Confidence Threshold curve.

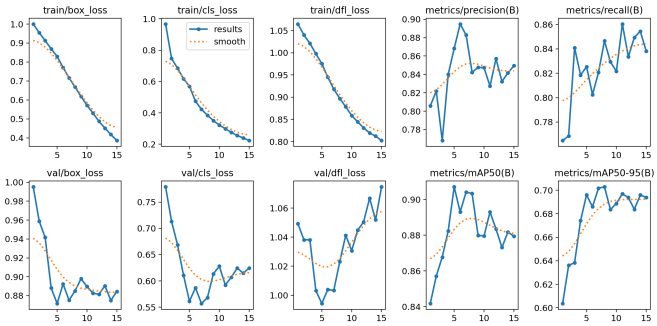

Fig. 11: Training and validation loss curves, along with precision trends.

F. Overall Evaluation Summary

TABLE IV: Precision, Recall, and mAP@0.5 metrics across varying thresholds

Precision (B)	Recall (B)	mAP50 (B)
0.80567	0.76483	0.8418
0.82193	0.76851	0.85711
0.76810	0.84087	0.86775
0.84002	0.81843	0.88243
0.86825	0.82539	0.90706

These results reaffirm the YOLOv8s model's effectiveness in detecting brake lights across a wide range of conditions, maintaining high precision with acceptable recall, even under occlusion and environmental variability.

V. Conclusion and Future Scope

This study demonstrates that the YOLOv8s model is an effective solution for detecting brake lights on four-wheeled vehicles. The training and validation loss curves indicate successful learning and strong generalization to unseen data. Evaluation metrics—including precision, recall, and mean Average Precision (mAP@0.5)—consistently show that the model achieves a favorable balance between detection accuracy and reliability, with an mAP close to 0.8 [12], [15].

However, it is important to acknowledge that the dataset used in this work primarily included standard passenger vehicles such as sedans, hatchbacks, and SUVs [25]. As a result, the current model was trained and evaluated exclusively on four-wheeled vehicles. The performance and generalizability of the model for other vehicle types, such as motorcycles, buses, and trucks, remain untested [14], [19]. This limitation should be considered when interpreting the applicability of the model in diverse traffic environments [5], [8].

Future Work

To further enhance the robustness and real-world usability of the proposed system, the following avenues are recommended for future exploration:

- **Expanding Dataset Diversity:** Increasing the dataset size and incorporating a broader spectrum of vehicle types—including two-wheelers and heavy vehicles—as well as varied environmental conditions (e.g., fog, rain, nighttime) would improve the model's generalization [5], [8], [13], [6].
- **Exploring Larger YOLOv8 Variants:** While YOLOv8s offers a favorable trade-off between speed and accuracy, future work could evaluate the performance of larger variants such as YOLOv8l and YOLOv8x to determine if higher accuracy can be achieved, particularly in complex or high-resolution settings [12], [2].
- **Integration of Vehicle Classification:** Introducing a multi-task model that simultaneously detects brake lights and classifies the type of vehicle could provide more context-aware detection, thereby improving both accuracy and application utility in ADAS systems [18], [24], [18].
- **Reducing False Positives and Enhancing Post-Processing:** Investigating techniques such as confidence threshold tuning, non-maximum suppression refinement, or the use of temporal consistency in video frames may reduce false positives and improve overall system reliability [16], [23], [26].
- **Benchmarking Against Transformer-Based Models:** Comparing the YOLOv8 model's performance with state-of-the-art transformer-based detectors like DETR and DINO could yield valuable insights into trade-offs between accuracy, speed, and computational efficiency in brake light detection tasks [28].

By addressing these directions, future studies can significantly enhance the adaptability and performance of deep learning-based brake light detection systems, supporting broader deployment in real-world autonomous and semi-autonomous driving scenarios [20], [21], [22], [9], [7], [17], [4], [10].

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